Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/04/2005
PI Name:	Shen, Hayley H Ph.D.		
Project Title:	Constitutive Relation in Transitional Granular Flows		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Granular flows		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	13699-5710	**Congressional District:**	23
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	02/13/2002	**End Date:**	12/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hanes, Daniel M. (University of Florida/Hanes and Associates)		
Grant/Contract No.:	NAG3-2717		
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Performance Goal Text:			
	The behavior of granular materials spans the whole range from an elastic solid to a non-linear viscous fluid. The constitutive relation depends not only on the material composition, but also on the deformation and the rate of deformation. The key reason for this behavior is that granular materials are not only defined by the properties of the individual grains, but also by the structure of the aggregate. In a very dilute granular flow, the structure of the aggregate is simply defined by a uniform random distribution. As the concentration increases, the internal structure becomes complex. It evolves dynamically with the macroscopic deformation and is an integral part of the constitutive relation. The complexity and importance of the internal structure increase with the concentration. In a gravity field, the weight of a granular material naturally compacts it to produce a concentration gradient. To study the constitutive behavior of granular materials, microgravity experiments must be performed to construct a physical basis for transitional constitutive		

Task Description:	relations. For long-term exploration on the Moon and the Mars, rheological properties of the surface granular materials require a better understanding. Moreover, many industrial problems on Earth deal with processing granular materials. Improving fundamental understanding of granular materials will also impact on industrial processes. This project combines both numerical and physical experiments to study the transition of granular flows from solid to gas phases. Using these experiments we will obtain data to quantify the internal resistance of a deforming granular material, to define and measure the internal structures, and to formulate theories to describe the transitional behavior. This study will achieve the following objectives: 1. A “phase diagram” will be established to characterize the transition of granular materials from solid to gas. 2. Physical parameters affecting the transition from one phase to another will be identified. 3. Internal structures of granular flows will be defined and measured. 4. The relation between the internal structures and the transition between phases will be identified. 5. An overall rheological theory for granular materials will be formulated. Flowing granular materials are commonplace in numerous terrestrial processes: landslides, sediment transport, and pneumatic transport of particulate solids in various manufacturing systems. Engineering designs for controlling these flows have relied heavily on empirical data. To date, while humans are capable of launching spacecrafts to land on Mars, failures of grain silos are still an unsolved problem, not to mention large-scale catastrophic granular flows such as debris avalanches. Unlike simple solids or fluids, a granular material under load can display deformation characteristics that cover the entire range from an elastic solid to a variety of non-linear viscous fluids. Many parameters affect the stress, strain, and strain-rate relation: particle size, shape, chemical composition, concentration, non-uniformity, humidity, and even the degree of deformation and rate of deformation. Our study is designed to understand the fundamental mechanisms that govern the rheological behavior of the granular materials so that better engineering designs can be made to solve problems with such materials on Earth as well as other celestial bodies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Improving the fundemental understanding of the rheological properties of granular materials will impact industrial processes such as pharmaceutical powder processing and design of construction (i.e. earth moving) equipment.
Task Progress:	In this year (January – December, 2004) our accomplishments are: 1. Completed the 3-D code. 2. Completed the physical testing with glass beads and Delrin beads. 3. Compared the results of the above two. 4. Made progress in theoretical development of transitional flows. 5. Acquired a Couette shear cell from Kevin Craig of Rensselaer Polytechnic Institute. The acquisition of the Couette shear cell is a big plus for the group. This addition makes Clarkson the only one in the US where all three essential elements in studying granular materials are co-located: theory, computation, and physical experiment. The PI, HH Shen, coPIs, DM Hanes and JT Jenkins met as a group in October to go over issues in data comparison from physical and numerical tests. HH Shen and JT Jenkins also met in January, March, and twice in November to discuss the theoretical issues. Transition is a difficult phenomenon, many researchers in the world, including both physicists and materials engineers are studying it. In addition, Dr. M Curt Koenders of Kingston University UK came to visit for 2 weeks in November. He gave two lectures, one in vibrations assisted infiltration process, the other in micromechanics of critical state soil mechanics. Dr. Koenders, JT Jenkins and the PI spent much time discussing the theory of granular transition. Dr. Koenders was the officemate of the PI last year at Newton Institute in Cambridge UK.
Bibliography Type:	Description: (Last Updated: 02/13/2008)
Articles in Other Journals or Periodicals	Ji, S.; Shen, H. H. "Contact force models for granular flows" Clarkson University Civil and Environmental Engineering departmental report, Report No. 04-02 , Mar-2004
Articles in Other Journals or Periodicals	Ji, S.; Shen, H. H. "Basic study on simple shear flow dynamics with periodic boundary condition in 3-D" Clarkson University Civil and Environmental Engineering departmental report, Report No. 04-05 , Apr-2004
Articles in Peer-reviewed Journals	Shen, H. H.; Sankaran, B. "Internal length and time scales in a simple shear granular flow" Phys. Rev. E , Nov-2004
Presentation	Shen, H. H.; Jenkins, J. T.; Ji, S. "Geometrically constraint flow behavior with uniform disks" APM04 (XXXII International Summer School – Conference “Advanced Problems in Mechanics”), St. Petersburg, Russia, June 24-July 1, 2004 Jul-2004
Presentation	Shen, H. H.; Hanes, D. M.; Ji, S. "Direct comparison of Couette shear cell tests between physical and computer experiments" ASCE 17th Engineering Mechanics Conference, U. Delaware, June 13-16, 2004 Jun-2004
Presentation	Shen, H. H. "Towards a Hydrodynamic Model for Granular Flows" Dalian University of Technology, Engineering Mechanics Department Jul-2004